Building Electricity Consumption Prediction Using Linear Regression and Support Vector Regression with Lag-Based Feature Engineering

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***Abstract*—Accurate prediction of building electrical energy consumption is critical for energy management and demand-response programs. This study presents an end-to-end machine-learning pipeline for short-term appliance-energy prediction using the UCI *Appliances Energy Prediction* dataset (19,735 ten-minute observations). Linear Regression (LR) and Support Vector Regression (SVR) with an RBF kernel are evaluated on 80:20 and 70:30 splits with 5-fold cross-validation. Comprehensive feature engineering (cyclical encoding, lag features on the log-transformed target, rolling statistics, and interactions) raises the LR R^2 on the original scale from 0.1694 (raw) to 0.5739 (+238.9%). After hyperparameter tuning, the SVR-RBF model with $C = 0.5$, $\gamma = 0.01$, $\epsilon = 0.2$ attains $R^2 = 0.7181$ on the log scale (0.5578 on original), MAE = 28.03 Wh, RMSE = 69.80 Wh, MAPE = 22.24%, competitive with prior gradient-boosting work on the same dataset. An inference experiment on 30 unseen test samples shows that 86.7% of predictions fall within ± 50 Wh of ground truth (median absolute error 13.7 Wh). Four practical use cases for the building operator are proposed.**

***Index Terms*—energy prediction, regression, support vector regression, feature engineering, building energy management, IoT, machine learning**

I. Introduction

Electrical energy management is a strategic priority for commercial-building operators because appliance use accounts for a significant share of operating costs. Accurate short-term forecasting enables demand-response decisions, peak-load shaving, and the early detection of equipment anomalies [2], [3]. The proliferation of Internet-of-Things (IoT) sensors makes fine-grained environmental data readily available, motivating data-driven prediction approaches in place of static rule-based controllers.

This study addresses the operational case of *PT. Cahaya Nusantara Energy*, a building-management company that operates office buildings and shopping centers across Indonesia. Using the publicly available *Appliances Energy Prediction* dataset [1] from the UCI Machine Learning Repository, we develop a regression pipeline that predicts the total appliance energy consumption (Wh) at 10-minute intervals from environmental sensor measurements and temporal context. Our contributions are: (i) a complete end-to-end ML pipeline covering EDA, feature engineering, baseline modeling, cross-validation, and tuning, in line with established practice [6]; (ii) a comparative evaluation of LR and SVR [4], [5] on two splits; (iii) quantitative analysis of how lag-based feature engineering and log-target transformation improve baseline performance; (iv) an inference experiment on unseen test samples; and (v) practical recommendations for deployment.

II. Methodology

A. Dataset

The dataset comprises 19,735 records collected at 10-minute intervals between January 11 and May 27, 2016 (137 days) from a low-energy house in Stambruges, Belgium. It contains 28 features besides the target: indoor temperature (T_1–T_9) and humidity (RH_1–RH_9) in nine zones, outdoor weather (temperature, humidity, pressure, wind, visibility, dew point), lighting consumption, and two intentional random-noise variables (`rv1`, `rv2`) inserted by the original authors to indicate the intrinsic noise floor [1]. The dataset has no missing values and no duplicate timestamps. The target `Appliances` (Wh) exhibits strong positive skew (skewness ≈ 3.39).

B. Feature Engineering

The Pearson correlation heatmap (Fig. 1) reveals strong intra-zone temperature correlations (above 0.85) and a near-perfect correlation between T_6 and T_{out} (above 0.95), motivating feature selection. Temporal exploration confirms clear daily peaks at $\sim$11:00 and $\sim$18:00 with reduced weekend consumption.

Six classes of features are engineered (27 new features in total): (i) calendar features (`hour`, `dayofweek`, `month`, `is_weekend`); (ii) cyclical $\sin / \cos$ encoding of hour, day-of-week, and month, which preserves natural periodicity (e.g., hour 23 and hour 0 are adjacent); (iii) lag features on the log target $\log y_{t-k}$ for $k \in \{1, 2, 3, 6, 12, 24, 144\}$, where the lag of 144 corresponds to one day; (iv) rolling mean and standard deviation of the log target over windows $\{3, 6, 12, 24\}$, shifted by one to prevent target leakage; (v) two first-difference features of the log target; and (vi) two interaction terms, $T_{\mathrm{out}} \cdot \sin(\mathrm{hour})$ and $T_{\mathrm{out}} \cdot \cos(\mathrm{hour})$.

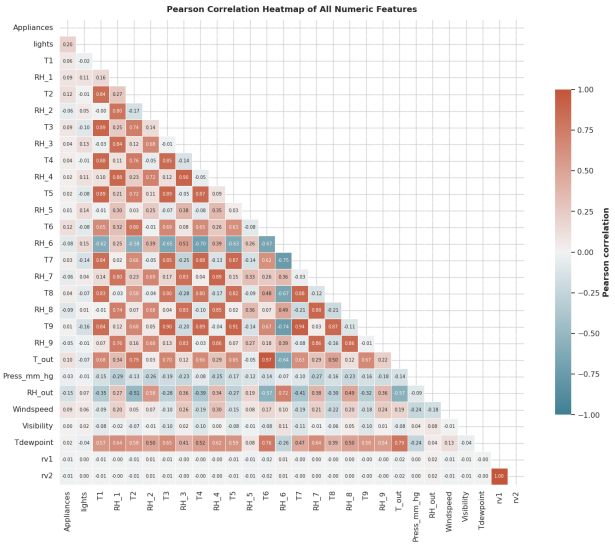

Fig. 1: Pearson correlation heatmap of the raw features. Indoor temperature variables exhibit strong multicollinearity.

A logarithmic transformation $y' = \log(1 + y)$ reduces target skewness from 3.39 to 0.69. After feature engineering and removal of 144 NaN rows produced by lag and rolling operations, 19,591 records remain. One pair with $|r| > 0.95$ is identified, leading to the removal of `T_out` and `ylog_rm_3`; 50 final features are retained. All inputs are z-score standardized, with the scaler fitted only on training partitions to prevent leakage.

The combined effect is substantial: on the 80:20 split, the original-scale R^2 of LR rises from 0.1694 (raw, original-scale target) to 0.5739 (engineered + log target), representing a +238.9% relative gain, while the MAE drops from 52.55 Wh to 28.23 Wh.

C. Models, Evaluation, and Tuning

LR fits $\hat{y} = \mathbf{w}^\top \mathbf{x} + b$ by ordinary least squares. SVR with the RBF kernel $K(\mathbf{x}, \mathbf{x}') = \exp\!\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right)$ is governed by hyperparameters C, γ, and ϵ [4], [5]. Because of SVR's $\mathcal{O}(n^2)$ to $\mathcal{O}(n^3)$ training cost, we train it on a random subsample of 5,000 observations, a methodological compromise acknowledged as a study limitation. Performance is reported in R^2 (log and original scales), MAE, RMSE, and MAPE. Each model is evaluated on the 80:20 and 70:30 splits and via 5-fold cross-validation. For tuning, Ridge uses 5-fold Grid Search over $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. SVR uses 3-fold Randomized Search with 15 sampled configurations on a sub-sample of 3,500 observations, drawn from $C \in \{0.5, 1, 2, 5, 10, 20, 50, 100\}$, $\gamma \in \{\text{scale, auto}, 10^{-3}, 5 \cdot 10^{-3}, 10^{-2}, 5 \cdot 10^{-2}, 10^{-1}\}$, and $\epsilon \in \{0.01, 0.05, 0.1, 0.2\}$. Randomized Search explores high-dimensional spaces more efficiently than Grid Search [7].

III. Results and Discussion

A. Baseline Performance and Cross-Validation

Table I reports baseline performance. LR and SVR achieve very similar R^2 on the log scale (≈ 0.71). LR slightly outperforms SVR on the original-scale metrics, while SVR yields lower MAPE values, reflecting better point-wise relative accuracy. The 70:30 split yields marginally better metrics due to its larger test set; differences are within sampling noise. The 5-fold cross-validation confirms model stability: LR yields $R^2 = 0.7107 \pm 0.0085$ and SVR yields $R^2 = 0.7114 \pm 0.0084$ on the log scale, with cross-fold standard deviation below 0.01 indicating consistent generalization.

TABLE I: Baseline Performance Across Two Train-Test Splits

Model	**Split**	R^2 **(log)**	R^2 **(orig)**	**MAE**	**RMSE**
LR	80:20	0.7123	0.5739	28.23	68.51
LR	70:30	0.7171	0.5843	27.38	66.39
SVR (RBF)	80:20	0.7092	0.5414	28.76	71.09
SVR (RBF)	70:30	0.7118	0.5444	27.95	69.51

TABLE II: Performance Before vs. After Hyperparameter Tuning (80:20)

Configuration	R^2 **(log)**	R^2 **(orig)**	**MAE**	**RMSE**
LR (vanilla)	0.7123	0.5739	28.23	68.51
Ridge ($\alpha = 100$)	0.7124	0.5729	28.23	68.60
SVR (default)	0.7092	0.5413	28.76	71.09
SVR (tuned)	**0.7181**	**0.5578**	**28.03**	**69.80**

B. Effect of Tuning and Best-Model Selection

Table II compares performance on the 80:20 split. Ridge tuning ($\alpha = 100$) yields negligible gains over vanilla LR, while SVR tuning ($C = 0.5, \gamma = 0.01, \epsilon = 0.2$) significantly improves the baseline, achieving the best performance: $R^2 = 0.7181$ (log), $R^2 = 0.5578$ (original), $\mathrm{MAE} = 28.03$ Wh, $\mathrm{RMSE} = 69.80$ Wh, $\mathrm{MAPE} = 22.24\%$, and CV $R^2 = 0.7257$.

The tuned SVR-RBF is selected as the final model due to its superior log-scale R^2, CV score, and minimal error metrics. The RBF kernel effectively captures non-linear interactions between lag features and temporal variables that linear models fail to represent. Despite marginal LR advantages on the 70:30 split, SVR's consistent log-scale performance justifies its use for peak-demand operations. Ridge-coefficient analysis identifies recent past consumption as the primary predictor, dominated by `ylog_lag_1`, `ylog_d1`, and log-target rolling means.

C. Diagnostic Analysis

Fig. 2 presents seven diagnostic panels. The actual-vs-predicted scatter plots in (a)–(b) show good alignment with the identity line. Residuals are symmetric and approximately Gaussian (c), with no systematic trend versus predicted values (d), satisfying the homoscedasticity assumption. The Q–Q plot (e) is nearly linear, confirming approximate normality. Panel (f) confirms an MAE of about 28 Wh. Panel (g) shows that the model tracks the overall pattern faithfully but underestimates extreme peaks, a limitation quantified per consumption segment in Table III.

For 78% of test observations (idle/low states), the model achieves high accuracy (max MAE 13.78 Wh). However, errors rise sharply in high/peak segments, reflecting the difficulty of predicting sporadic events driven by stochastic occupant behavior rather than environmental conditions.

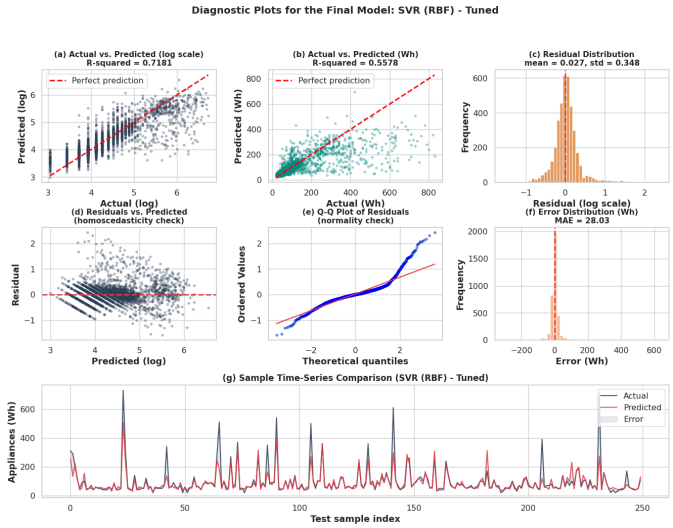

Fig. 2: Diagnostic plots of the final model (SVR-RBF tuned): (a)–(b) actual vs. predicted on log and original scales; (c) residual distribution; (d) residuals vs. predicted; (e) Q–Q plot; (f) error distribution in Wh; (g) sample time-series of 250 test points.

TABLE III: Per-Segment Error Analysis on the 80:20 Test Set

Segment (Wh)	n	**Actual**	**Pred.**	**MAE**	**MAPE (%)**
<50 (idle)	1457	43.61	49.10	8.59	22.36
50–100 (low)	1601	73.60	75.18	13.78	18.24
100–200 (medium)	473	132.33	128.68	31.61	22.81
200–500 (high)	317	315.68	221.56	111.35	33.39
>500 (peak)	71	618.73	266.66	352.07	56.50

D. Inference on Unseen Test Samples

To assess deployment readiness, an inference experiment on 30 unseen samples demonstrated strong accuracy in the typical operating range: 86.7% (26/30) of predictions fall within ± 50 Wh of ground truth, with a median absolute error of 13.74 Wh. While the mean absolute error is 52.92 Wh due to the under-prediction of sporadic high-magnitude peaks (e.g., actual values of 640 and 480 Wh predicted at 148 and 147 Wh), the model remains reliable for routine operational forecasting.

E. Comparison, Use Cases, and Limitations

The original-scale $R^2 = 0.5578$ is competitive with the $R^2 \approx 0.57$ reported by Candanedo *et al.* [1] using Gradient Boosting. This parity, achieved using simpler baseline models, highlights the effectiveness of robust feature engineering and target transformation. We propose four applications for PT. Cahaya Nusantara Energy: (i) *short-term load forecasting* (10 to 60 minutes ahead) for demand response, with estimated 5 to 10% savings via load-shedding; (ii) *anomaly detection* for maintenance by flagging intervals where $|y_t - \hat{y}_t| > 3 \cdot \mathrm{MAE}$; (iii) *HVAC and lighting optimization* for peak-shaving (estimated 10 to 15% savings); and (iv) *monthly budget forecasting* to support contract negotiations. Study limitations include the use of a 5,000-sample SVR training subset due to $\mathcal{O}(n^2)$ to $\mathcal{O}(n^3)$ computational costs, and the model's peak-event under-prediction, which necessitates pairing regression with a dedicated peak-detection module for reliable deployment.

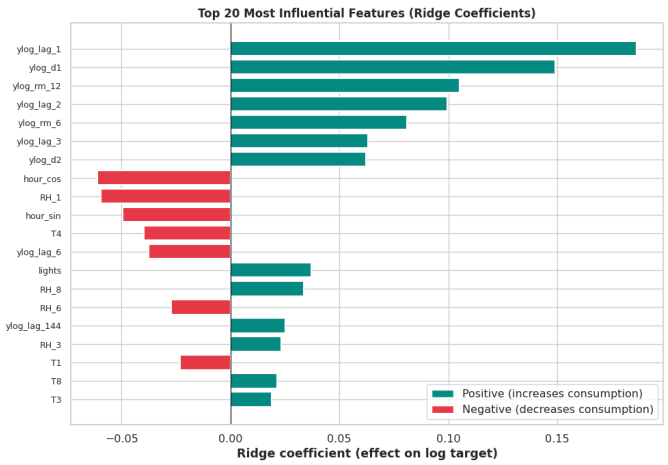

Fig. 3: Inference on 30 random unseen test samples: (a) actual vs. predicted bars per sample; (b) scatter alignment with the identity line and the ± 50 Wh tolerance band. 86.7% of predictions fall within the tolerance; median absolute error = 13.7 Wh.

IV. Conclusion

This study developed an effective machine-learning pipeline for appliance energy prediction, where the tuned SVR-RBF model achieved an $R^2 = 0.7181$ (log) and $R^2 = 0.5578$ (original), with $\mathrm{MAE} = 28.03$ Wh, $\mathrm{RMSE} = 69.80$ Wh, and $\mathrm{MAPE} = 22.24\%$. Inference testing showed that 86.7% of predictions fall within ± 50 Wh of ground truth, confirming the model's reliability despite error concentrations in peak segments. Lag features and rolling statistics proved to be the primary predictors of consumption. Future research will investigate tree-based ensembles and recurrent neural networks (RNNs) [8] to better capture non-linear, long-term temporal dependencies and integrate occupancy data to reduce stochastic noise.

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